

Trustworthy and Explainable Agents for Phishing Detection with Grounded Cross-Model Evidence Agreement

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Abstract

When a phishing detector marks a website, the verdict drives a browser warning, a blocklist entry, or a take-down request, so the operator who acts on it needs a reason they can check and a rule for when the verdict is safe to trust on its own. Large language models now reach high accuracy here, yet they emit a *label* and discard the *reason*, and the usual reliability signals – model confidence, self-consistency, and majority voting – read trust at the label, where several models can agree on the right verdict while relying on different or even fabricated evidence. To close this gap, we propose PhishProof, a selective phishing detector that reads trust from grounded cross-model evidence agreement instead of from labels. A small panel of diverse language-model agents investigates each page, plans which of its URL, DOM, certificate, redirect, screenshot, and logo signals to gather, and calls verification tools – including existing reference-based and LLM detectors – to confirm every claim against the page itself. PhishProof acts only when the agents agree on cues the page confirms and otherwise abstains, returning those agreed, verified cues as a checkable explanation when it does act; a held-out labeled split calibrates the agreement-and-groundedness score to a probability of correctness and sets the operating point. Because the panel calls specialized detectors as tools, it stays competitive with them on full-coverage accuracy while adding the calibrated abstention and verifiable explanation they lack. On a phishing benchmark drawn from public feeds and the Tranco top sites, PhishProof ranks risky verdicts more reliably than the strongest label-agreement baseline, acts confidently on far more pages at a fixed high selective accuracy, and abstains on most of the right-verdict-but-wrong-reason and ungrounded cases that baselines wave through.

Keywords: phishing detection, LLM agents, tool-augmented agents, evidence cues, selective prediction, cross-model evidence agreement, explanation faithfulness, abstention, trustworthy ML

1. Introduction

A phishing verdict is a consequential act: marking a page as phishing drives a blocklist entry, a browser warning, or a take-down request that pulls a live service off the web. Large language models have sharply raised the accuracy of such detectors, reading a page’s URL, DOM, rendered screenshot, and certificate to decide whether it impersonates a brand it does not own [1, 2]. Yet accuracy is only half of what a deployment needs: the operator who must justify a take-down and the user who sees a warning both need a *reason* they can check, and the system needs a criterion for *when* a verdict is trustworthy enough to act on without review. For a decision that can be contested, the bottleneck is no longer whether the model is usually right, but whether a particular verdict can be *trusted and explained*.

Existing detectors approach this only indirectly. One line runs a single LLM over the page and reports a class probability or a verbalized confidence [1]; reliability is then a property of that one model’s calibration [3]. A second line aggregates

several models or several samples – self-consistency over one model [4], majority voting over an ensemble [5], or weighted label agreement in the style of crowd-sourcing and label-free evaluation [6, 7] – and reads reliability off how often the votes concur. A third, recent line builds multi-agent detectors whose agents specialize on different signals and whose rationales are merged into a single explanation [8, 9]. Together these families cover much of the space of accuracy and aggregation.

They share a blind spot, visible in a routine case (Fig. 1). A page impersonating a bank is shown to a panel of three diverse models; all three return “phishing,” so every aggregation method above reports high confidence. But the models reached that verdict for different reasons: one cited the morphed logo, one cited a credential form posting off-domain, and one cited a certificate mismatch that does not actually hold on the page. First, the detectors emit a label and discard these reasons, so nothing auditable reaches the operator (**L1**). Second, reliability is read at the *label*, where the three agree, and is therefore blind to the fact that they *disagree on the evidence*: the verdict is right for the wrong reason (**L2**). Third, even where the models do agree on a reason, that agreement can be a shared hallucination of a cue that is not on the page (**L3**). A right-label, wrong-reason verdict is not a curiosity: it is precisely the

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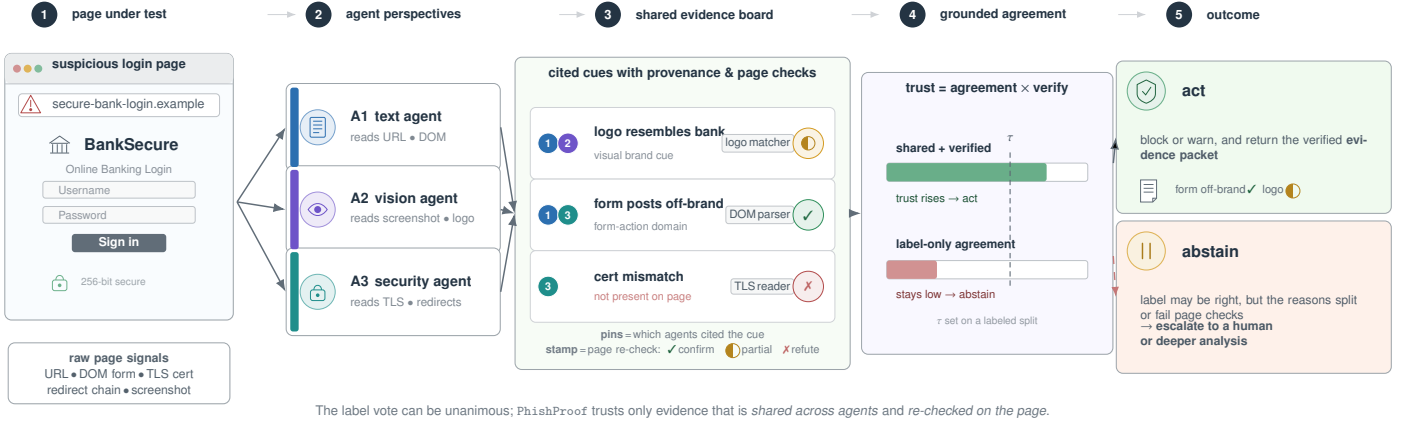


Figure 1: PhishProof as a qualitative evidence story. Three agents inspect the same page from complementary views, then place their cited cues on a shared evidence board. Colored pins show provenance, while tool badges mark cues that are confirmed or rejected by the page itself. PhishProof acts when the shared reasons survive these checks and returns them as an evidence packet; when agents agree on the label but their reasons split or fail verification, it abstains.

verdict a small change to the cited cue would flip, and precisely the one an operator should not act on blindly.

Two properties of phishing, not shared by generic question answering, make a direct remedy possible. First, the evidence a detector relies on is *objectively checkable*: a form-action domain, a certificate organization, a redirect target, and a rendered logo can each be re-derived from the page itself. Second, labeled pages are cheap to assemble – public feeds publish freshly confirmed phishing URLs and the Tranco list [10] supplies benign sites – so a detector’s operating point can be set on a held-out labeled split rather than guessed. The evidence that decides a verdict is also *heterogeneous*, spanning several model types (text LLMs and a vision-language model), many signals (URL, DOM, certificate, redirect chain, logo, and credential intent), two modalities, and a verification mechanism per signal. Deciding which signals to gather for a given page and how to combine them is a *planning-and-aggregation* problem better suited to an agent than to a fixed pipeline, and modern LLMs can act as such agents – emitting typed outputs and calling tools to gather and verify their own evidence. The opportunity is to stop treating reliability as a property of labels and start treating it as a property of *evidence that several independent agents plan, gather, verify, and agree on, and that the page confirms*.

To this end, we present PhishProof, a multi-agent phishing detector that trusts a verdict in proportion to grounded cross-model agreement on the evidence. A small, deliberately diverse panel of LLM *evidence agents* investigates the page: rather than emit a bare label, each agent *plans* which signals and modalities to examine, then gathers and grounds its evidence by calling verification *tools* – a DOM parser, a certificate reader, a redirect tracer, a logo matcher, and an existing reference-based or LLM brand detector – and records what it found in a shared, per-page set of typed *evidence cues*, each carrying the asserting agent and the confirming tool. PhishProof *aggregates* these grounded findings: it measures agreement at the level of the cited *evidence* rather than the label, weights that agreement by whether each agreed cue verifies against the page, and combines the two into a single Grounded Evidence-Agreement score. It acts only

when the score is high – returning the agreed, verified cues as a checkable explanation – and otherwise abstains, escalating the page to deeper analysis or a human. A calibrator fit on a held-out labeled split maps the score to a probability of correctness, so the operating point is set against real outcomes rather than guessed. Reusing the existing detectors as tools also makes PhishProof competitive on full-coverage accuracy by construction, since the panel inherits their signal. The novelty over prior multi-agent detectors is the unit of agreement: not the label, and not one model’s introspective faithfulness, but a set of evidence cues that is simultaneously cross-agent-agreed and tool-verified.

Realizing this raises three difficulties that the design must rule out. First, to answer **L2**, agreement over structured evidence must be defined so that it cannot collapse into label agreement in disguise – two models can name the same brand while resting on entirely different cues. Second, to answer **L3**, shared hallucination must be caught cheaply, by checking cited cues against the page rather than trusting the models to police themselves. Cutting across both, the panel must stay small: every added agent is another round of model and tool calls, so the gain has to come from agent *diversity* rather than count.

We make the following contributions.

- **Failure mode.** We identify the *right-label, wrong-reason* failure of multi-model phishing detectors and cast trustworthy detection as selective prediction over an evidence-agreement signal (§3).
- **Method.** We introduce PhishProof, a panel of tool-using evidence agents that cite a shared set of typed, verified evidence cues, and its Grounded Evidence-Agreement score, which measures cross-agent agreement over those cues and turns it, calibrated on a held-out labeled split, into a selective act/abstain decision with the agreed, verified cues as the explanation (§4).
- **Evidence.** On a phishing benchmark assembled from public feeds and the Tranco top sites, PhishProof is competitive with specialized reference-based and LLM detectors

on full-coverage accuracy – it invokes them as verification tools – and improves the risk–coverage trade-off over single-model confidence, self-consistency, and label voting: AURC falls by 35% relative to the strongest baseline and coverage at 99% selective accuracy rises from 0.48 to 0.71 (§5).

- **Analysis.** We isolate the contribution of evidence-level agreement, tool-grounding, agent diversity, and calibration, and show the failure modes are real: the right-label, wrong-reason and ungrounded regions flip under evidence-targeted perturbations 41.6% and 47.9% of the time, and PhishProof abstains on 78–83% of them (§5).

The rest of the paper is organized as follows. §2 positions PhishProof against label-level reliability and multi-agent detection. §3 formalizes grounded evidence-agreement and the selective objective. §4 details the evidence agents, the cues they cite, the score, and the calibrated decision rule, and §5 reports the evaluation.

2. Related Work

LLM and reference-based phishing detection. The dominant modern detectors decide whether a page impersonates a brand it does not own. Reference-based systems match a rendered page against a brand library of logos and layouts [11, 12, 13], and recent work refreshes that reference set online [14] or replaces it with an LLM that recognizes the brand and checks the domain directly [1, 2]. A parallel line treats the page as URL or image features for a learned classifier [15], surveyed broadly [16]. Across these sub-mechanisms the shared target is *aggregate* accuracy, reported as a single page-level verdict. Building on this line rather than competing with it, PhishProof invokes such detectors as verification tools (§4.3) – inheriting their accuracy so that it stays *competitive* at full coverage (Tab. 3) – and adds what a bare detector lacks: a *per-verdict* decision about whether the verdict can be trusted, acted on automatically, and explained (§4.4).

Multi-agent and ensemble phishing detection. Closest to us are detectors that run several models. Majority voting over LLMs has been studied directly for phishing [5]; multi-agent systems assign agents to different signals and either debate to a verdict or merge their rationales into one explanation [8, 9], sometimes backed by an episodic memory of past cases [17]. Unlike these systems, which aggregate or consolidate at the level of the *label* or a fused rationale, PhishProof measures agreement at the level of individually cited evidence units ((2)) and treats *disagreement on evidence* as the signal to abstain rather than a conflict to resolve. Its agents are tool-using investigators that each *plan* their own investigation across signals and modalities, and whose tool-grounded findings are *aggregated* (§4.3) into a shared set of cited evidence cues that serves as both the trust signal and the explanation, rather than a fused free-text rationale.

Label-level reliability and label-free evaluation. A verdict’s reliability is usually read from the label distribution:

post-hoc confidence calibration [3], agreement across samples of one model [4], or agreement across many models, as in crowd-sourcing aggregation [6], agreement-on-the-line under shift [18], and trust calibration for phishing detectors [7]. Where these estimate reliability from *how often labels concur*, PhishProof estimates it from *whether the reasons concur and hold*, which separates pages on which the panel agrees by coincidence from those on which it agrees because the page truly carries the evidence ((4)).

Explanation faithfulness. Causal audits show that an LLM’s stated rationale is often not the driver of its decision [19], and multi-agent debate has been used to *improve* the faithfulness of generated explanations [20]. Such audits operate on a single model and require counterfactual re-execution. In contrast, PhishProof uses cross-model agreement on grounded evidence as an inexpensive faithfulness *proxy*: a reason that several independent models cite and that the page confirms ((3)) is, by construction, less likely to be a post-hoc rationalization, at no extra re-execution cost.

Selective prediction. We build on selective classification, which trades coverage for accuracy via an abstention rule [21], and on distribution-free uncertainty quantification [22]. Building on this foundation, PhishProof supplies the missing ingredient for the phishing setting: an abstention score derived from grounded cross-model agreement ((4)), calibrated on a held-out labeled split so the risk–coverage operating point is set against real outcomes (§4.4).

Position and scope. PhishProof contributes a single new object, the Grounded Evidence-Agreement score, and the selective detector built on it. Tab. 1 positions it against one representative per family along five capabilities – evidence-level agreement, page-grounding, calibrated trust, selective abstention, and a verifiable explanation as output – and PhishProof is the only entry that supports all five. We focus on per-verdict

Table 1: Capability comparison for per-verdict trust in multi-model phishing detection. ●: yes; ○: partial; ○: no. Columns are defined in the body; full-coverage detection accuracy is reported separately in Tab. 3.

Approach	Evid.-level	Page-grounded	Calib.-trust	Selective abstain	Explan. output
Phishpedia [12]	○	●	○	○	●
PhishLLM [1]	○	●	○	○	●
Single conf. [3]	○	○	●	●	○
Self-consist. [4]	○	○	○	●	○
Majority vote [5]	○	○	○	●	○
Label agree. [6]	○	○	●	●	○
MultiPhishGuard [9]	●	○	○	●	●
Faithfulness audit [19]	●	○	○	○	●
PhishProof (ours)	●	●	●	●	●

trust and explanation for a multi-model phishing detector; the framework is agnostic to the specific base models and evidence schema. We do *not* claim (i) a new base detector or a leaderboard win on full-coverage accuracy – PhishProof reuses existing detectors as tools and is *competitive* with them (Tab. 3), contributing the per-verdict trust and explanation layer on top; (ii) robustness to an adversary who controls the panel agents or

corrupts the tools’ inputs, a harness-level threat we treat as out of scope; or (iii) coverage guarantees on fully cloaked pages that carry no verifiable cues, where the detector necessarily abstains rather than errs.

3. Problem Formulation

We formalize trustworthy phishing detection as *selective prediction over a grounded evidence-agreement score*: rather than always emitting a label, the detector acts only when its evidence is both agreed-upon and verified, and otherwise abstains. We proceed in five steps – what the detector observes (§3.1), the score it computes (§3.2), the act/abstain objective it optimizes (§3.3), the requirements the method must meet (§3.4), and the resulting framework (§3.5). Tab. 2 collects the notation.

3.1. Preliminaries

Setting. A detector receives a webpage x – represented by its URL, DOM, rendered screenshot, TLS certificate, and redirect chain – and must decide whether x is phishing. It has access to two resources. The first is a panel $\mathcal{M} = \{m_1, \dots, m_M\}$ of M diverse LLM evidence agents: they come from different model families and include at least one vision-language model that reads the screenshot, and each can call a set of verification tools that re-derive factual signals from x . The second is a fixed, typed evidence schema $\mathcal{U}$ that fixes the vocabulary of facts the agents may cite. A cue $u = (t, v) \in \mathcal{U}$ pairs a type t (e.g. form-action-domain, cert-org, redirect-target, logo-brand, credential-intent) with a value v (e.g. the domain a login form posts to); every type has a corresponding tool g that re-derives it from x , which is what makes a cue checkable rather than merely asserted.

Agent outputs and the shared cue set. Run on x , each agent investigates the page and returns a verdict together with the cues it grounds through tool calls:

$$m_i(x) = (y_i(x), E_i(x)), \quad y_i \in \{\text{phish}, \text{benign}\}, \quad E_i \subseteq \mathcal{U}. \quad (1)$$

Pooling the agents’ cited cues gives the page’s *shared cue set* $\bigcup_i E_i(x)$, the universe of evidence anyone raised, and the verdict is the majority vote $\hat{y}(x) = \text{maj}_i y_i(x)$. Within the shared set, a *consensus* cue is one that a strict majority of the agents assert with a consistent value, $C(x) = \{u : |\{i : u \in E_i\}| > M/2\}$ – the evidence the panel independently corroborates; these consensus cues are the evidence PhishProof trusts to drive the verdict and returns as the explanation.

3.2. Grounded Evidence-Agreement

Existing detectors estimate per-verdict reliability from the *label* channel alone – a single model’s confidence, or the concordance of the verdicts $\{y_i\}$ [3, 5, 6]. The flaw is that the label channel is blind to *why* the agents agree. We instead score the *evidence* channel with two complementary factors: how much of the cited evidence the panel agrees on, and how much of that agreed evidence the page actually bears out. Formally, the

Table 2: Summary of important notations.

Symbol	Meaning
x	a webpage under test (URL, DOM, screenshot, cert, redirects)
$\mathcal{M} = \{m_1, \dots, m_M\}$	panel of M diverse LLM evidence agents ($M=3$)
$\mathcal{U}$	fixed typed evidence-cue schema (§4.1)
$E_i \subseteq \mathcal{U}$	grounded cues agent m_i cites on x
C	<i>consensus</i> cue set (cues a majority of agents assert)
$g(u, x) \in [0, 1]$	tool that re-derives cue u ’s type from x
$A(x) \in [0, 1]$	evidence-level agreement over the shared cue set
$G(x) \in [0, 1]$	groundedness of the consensus cues
$\text{GEA}(x) = A \cdot G$	Grounded Evidence-Agreement score
τ	abstention threshold (acts iff $\text{GEA} \geq \tau$)
$\hat{p} = \kappa(\text{GEA})$	calibrated map $\text{GEA} \rightarrow \mathbb{P}(\text{correct})$, fit on a labeled split

evidence-level agreement and the groundedness of the consensus are

$$A(x) = \frac{|C(x)|}{|\bigcup_i E_i(x)|}, \quad (2)$$

$$G(x) = \frac{1}{|C(x)|} \sum_{u \in C(x)} g(u, x), \quad (3)$$

where $g(u, x) \in [0, 1]$ is the tool that re-derives cue u ’s type from x (deterministic, returning 0 or 1, for form-action, cert-org, and redirect; perceptual, returning a similarity, for logo-brand), and we set $G(x) = 0$ when $C(x)$ is empty. Thus A rises only as the agents converge on the *same* cues, and G rises only as those agreed cues survive re-derivation from the page. The *Grounded Evidence-Agreement* score is their product,

$$\text{GEA}(x) = A(x) \cdot G(x) \in [0, 1]. \quad (4)$$

The product, rather than a sum or average, is deliberate: either failure alone drives GEA to zero. Agreement on cues the page does not confirm sends $G \rightarrow 0$, and a panel that concurs on the label while citing disjoint cues sends $A \rightarrow 0$; only when the agents agree on the evidence *and* the page bears it out does GEA stay high. This is exactly the conjunction the label channel cannot express.

3.3. Selective Objective

PhishProof is a *selective* detector: it acts with the verdict $\hat{y}(x)$ when $\text{GEA}(x) \geq \tau$ for a threshold τ , and otherwise abstains, emitting $\perp$ to escalate the page. A threshold induces two quantities – the *coverage* $\phi(\tau)$, the fraction of pages on which the detector acts, and the *selective risk* $r(\tau)$, the error rate on exactly those pages:

$$\phi(\tau) = \mathbb{P}[\text{GEA} \geq \tau], \quad r(\tau) = \mathbb{P}[\hat{y} \neq y \mid \text{GEA} \geq \tau], \quad (5)$$

with y the true label. Sweeping τ traces the risk–coverage curve $r(\phi)$, and a good score keeps this curve low everywhere – equivalently, it minimizes the area under it (AUCR). To turn the raw score into an actionable threshold, we fit a calibrator $\kappa : \text{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$ on a held-out labeled split that maps the score to a probability of being correct, and pick the operating point τ that realizes a target risk. The per-page *output* is then a verdict or an abstention, the calibrated trust $\hat{p} = \kappa(\text{GEA})$, and, when acting, the consensus cue set $C(x)$ as the explanation.

3.4. Design Requirements

Scoring the evidence channel rather than the label channel imposes three requirements, each repairing one limitation of label-centric reliability identified in §1. The framework (§3.5) discharges each through a named component, and §5 measures its effect.

(R1) *Emit structured, checkable evidence.* The label channel returns no auditable artifact, only the verdict $\hat{y}$, so an action cannot be justified or verified (answers **L1**; realized by agentic evidence elicitation, component C_1 , §4.1, and returned as the explanation throughout).

(R2) *Measure agreement at the evidence level.* Concordance read from the verdicts $\{y_i\}$ cannot distinguish a panel that agrees *because* the page carries the evidence from one that agrees on the label while resting on entirely different cues (answers **L2**; realized by evidence agreement A , component C_2 , §4.2; ablated in §5.5).

(R3) *Ground cited evidence to the page.* Evidence the panel cites in agreement may still be a *shared* hallucination, which agreement alone would reward; it must instead be checked against x (answers **L3**; realized by tool-grounding G , component C_3 , §4.3; ablated in §5.5).

This limitation–requirement–component thread runs through the paper: each C_i is developed in §4 and isolated by an ablation row in §5.5, so the contribution of every requirement is measured, not assumed.

3.5. Framework Overview

End-to-end, PhishProof turns a page into a trusted-and-explained verdict in four steps, drawn in Fig. 2 and developed one per subsection in §4; the four components C_1 – C_4 map one-to-one onto these steps. (1) **Investigate.** Each agent plans which signals to examine and calls the verification tools, emitting its verdict and a set of typed, tool-checkable cues into the shared cue set (C_1 agentic evidence elicitation, **R1**, §4.1). (2) **Aggregate on evidence.** Over the shared set we form the strict-majority consensus and measure how much of the cited evidence falls within it – the evidence-level agreement A – instead of counting how often the verdicts concur (C_2 , **R2**, §4.2). (3) **Verify against the page.** Each consensus cue is re-derived by its tool, and the groundedness G discounts any consensus that the page does not actually bear out, catching shared hallucinations (C_3 , **R3**, §4.3). (4) **Decide and explain.** The score $GEA = A \cdot G$ is calibrated on a labeled split and thresholded into an act/abstain decision; when it acts it returns the consensus cues as the explanation, and otherwise escalates the page (C_4 calibrated selection, §4.4). Because the score is a product, a verdict survives to action only when steps (2) and (3) both hold – precisely guarding against the right-label/wrong-reason and shared-hallucination failures the label channel misses.

4. The PhishProof Detector

PhishProof turns the Grounded Evidence-Agreement score of (4) into a calibrated, selective verdict through the four components C_1 – C_4 previewed in §3.5 and drawn in Fig. 2. The

four subsections below develop them in the order a page flows through the system – elicit evidence, agree on it, ground it, then decide – after which we give the end-to-end procedure, its guarantee, and its cost.

4.1. Evidence agents and the shared cue set

This component produces the per-agent outputs of (1). Each member of the panel is an *evidence agent*: prompted with the page, it *plans* which signals and modalities to examine, calls the verification tools to obtain them, and returns a verdict together with the grounded cues it found, drawn from the fixed schema $\mathcal{U}$ and emitted by constrained decoding as well-typed (t, v) pairs rather than free text (**R1**). Constraining the output to the schema is what makes the cue auditable: it is a typed claim a tool can later re-derive, not prose to be re-interpreted. The schema is small and auditable – six types covering the brand claim, the form-action domain, the certificate organization, the redirect target, credential intent, and the logo brand – and is shared by every agent, so their cues are directly comparable. The cues are pooled into the page’s *shared cue set*, where each cue is typed, tool-checkable, and tagged with the agent that asserted it, so the verdict rests on a list of cited, verifiable reasons – a brand claim paired with off-brand infrastructure – rather than an opaque score. Agent *diversity* is a design choice, not an afterthought: the three agents come from different model families and at least one is a vision-language model that reads the screenshot, so that *agreement* on a cue reflects independent corroboration rather than a shared prior the agents happen to hold (its value is isolated in §5).

4.2. Evidence agreement

This component scores the shared cue set of §4.1. To *aggregate* and compare cues across agents, two agents must be able to assert the “same” cue despite surface differences, so we first *normalize values*: domains are reduced to their registrable eTLD+1 and brands to a canonical lexicon entry, so that “paypal-secure.com/login” and “paypal-secure.com/signin” count as one form-action cue. On the normalized cues, the consensus set $C(x)$ collects those a strict majority of the agents assert, and the evidence-level agreement $A(x)$ of (2) is the share of all cues raised that fall in this consensus (**R2**). Agreement is therefore high only when the agents converge on the *same* evidence, and – unlike a label-vote concordance – is unmoved by their merely concurring on the verdict.

4.3. Tool-grounding

This component verifies the consensus cues from §4.2 against the page. Agreement alone cannot catch a *shared* hallucination – a cue the whole panel asserts but the page does not carry – so each consensus cue is re-derived from the page by the tool for its type, $g(u, x)$ (**R3**). The structural tools verify deterministically and return 0 or 1: the DOM is parsed for the form-action domain, the certificate for its organization, and the redirect chain for its final target. The perceptual tool (logo-brand) matches the rendered logo to the claimed brand and returns a

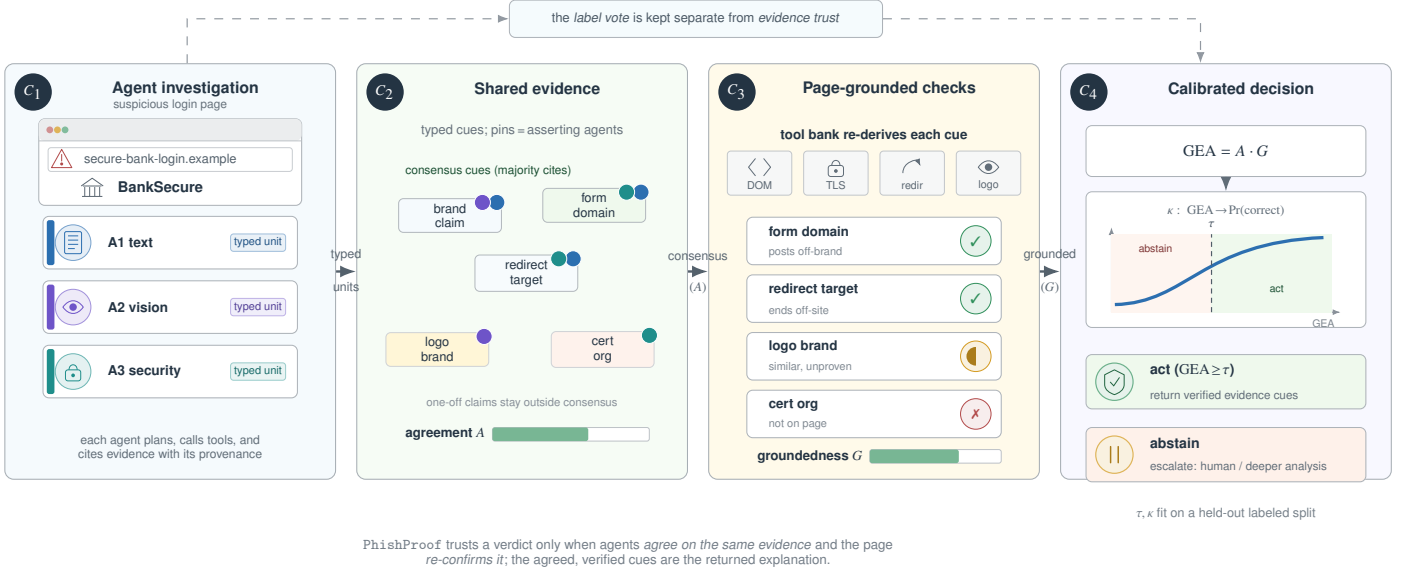


Figure 2: PhishProof qualitative pipeline. Agents inspect the same page from complementary views and cite typed evidence cues into a shared set. The system trusts a verdict only when the agents converge on the same cues and the tool bank can re-check those cues against the live page. The calibrated selector returns the label with the verified cues when the evidence is shared and grounded; otherwise it abstains, including cases where the label vote is plausible but the reasons are thin or unverified. The threshold and calibrator are set on a held-out labeled split.

similarity in $[0, 1]$. The brand claim itself is grounded by an existing specialized detector *used as a tool* – a reference-based visual matcher [12, 13] or an LLM brand-and-domain checker [1] – so the panel *subsumes* the signals of the dedicated phishing detectors rather than re-deriving them, and inherits their detection quality by construction (§5). The groundedness $G(x)$ of (3) averages these tool outputs over the consensus, and through the product $\text{GEA} = A \cdot G$ of (4) the score collapses whenever the agreed evidence does not survive verification.

4.4. Calibrated selection

Finally, PhishProof turns the score $\text{GEA}(x)$ into an act/abstain decision. Because phishing labels are available at collection time – a flagged URL is confirmed by the public feed⁴³⁰ it is drawn from [23, 24] – we can set the operating point on a small held-out labeled split rather than guess it. On that split we fit a monotonic (isotonic) calibrator $\kappa : \text{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$, which is order-preserving and so leaves the score’s ranking – hence the risk-coverage curve – intact while turning it into a⁴³⁵ probability of correctness, and we pick the smallest threshold τ that meets a target selective risk. The detector then acts with $\hat{y}(x)$ when $\text{GEA}(x) \geq \tau$ and abstains otherwise, escalating the page to deeper analysis or a human. When it acts, it returns the consensus cue set $C(x)$ as a verifiable explanation.

4.5. Procedure, guarantee, and cost

Walk-through. 1 runs the agent panel once, each agent calling its tools to cite its cues (line 1), takes the majority verdict (line 2), and then does the work that distinguishes PhishProof from label-level reliability: it forms the consensus set (line 3),⁴⁴⁵ scores agreement on that evidence rather than on the verdict (line 4), checks each agreed cue against the page with its tool

Algorithm 1 PhishProof inference on a page x

- 1: **for** $m_i \in \mathcal{M}$: $(y_i, E_i) \leftarrow m_i(x)$ ▷ C_1 agents call tools, (1)
- 2: $\hat{y} \leftarrow \text{maj}_i y_i$; $E \leftarrow \bigcup_i E_i$ ▷ shared cue set
- 3: $C \leftarrow \{u : u \text{ asserted by } > M/2 \text{ agents, value-normalized}\}$
- 4: $A \leftarrow |C| / |E|$ ▷ C_2 , (2)
- 5: **for** $u \in C$: compute $g(u, x)$; $G \leftarrow \text{mean}_{u \in C} g(u, x)$ ▷ C_3 , (3)
- 6: $\text{GEA} \leftarrow A \cdot G$; $\hat{\rho} \leftarrow \kappa(\text{GEA})$ ▷ (4)
- 7: **if** $\text{GEA} \geq \tau$ **return** $(\hat{y}, \hat{\rho}, C)$; **else return** $\perp$

so a shared hallucination cannot inflate the score (line 5), and combines the two factors into GEA (line 6). Calibration and the selective decision (line 7) then map the score to a trust value and apply the threshold fitted on the labeled split. The novelty is concentrated in lines 3–6: the quantity that gates the verdict is grounded agreement on the *cues*, not concordance of labels, and the returned consensus set is simultaneously the trust signal and the explanation – one object serving both **R2–R3** and the audit trail of **R1**.

Guarantee. Tool-grounding is the component that rules out a shared hallucination inflating the score, and under one assumption on the tools we can state this precisely.

Assumption 1 (Tool soundness). *For each cue type, the tool $g(u, x) = 1$ if u holds on x and 0 otherwise (perceptual types: $g(u, x)$ is the true match score).*

Proposition 1. *Under Assumption 1, any consensus cue that does not hold on x contributes 0 to $G(x)$; hence $\text{GEA}(x)$ is at most the grounded fraction of the consensus, and $\text{GEA}(x) = 0$ whenever the entire consensus is hallucinated.*

Proof sketch. G is the mean of g over C ((3)); by Assumption 1 a non-holding cue adds 0 to that mean, so $G \leq |\{u \in C :$

u holds $\|/|C|$, and $GEA = A \cdot G \leq G$ inherits the bound. If 500
no consensus cue holds, $G = 0$ and thus $GEA = 0$. $\square$

450 The proposition is modest by design: it bounds, but does not
eliminate, gauge error, since a tool is sound for *what* it checks
yet a page can still carry true-yet-misleading evidence the tool 505
confirms; accordingly we report measured tool precision and
recall against gold in §5 rather than rely on Assumption 1 as if
it held exactly.

460 **Complexity.** Per page, the cost has two parts: the M agent calls
of line 1 – each a model call plus the tool calls it issues – and the 510
verification of the $|C|$ consensus cues in line 5. The aggregation
steps (lines 2–4, 6) are linear in the number of cues raised and
negligible beside a model call. Each structural tool is an $O(1)$
DOM, certificate, or redirect lookup and the single perceptual
check is one embedding comparison, so verification is $O(|C|)$ 515
and the per-page cost is dominated by the $M=3$ agent calls;
it is independent of the schema size beyond the cues actually
asserted, and there is no training. Relative to the strongest non-
agentic baseline – a single-model detector – this is a fixed $M \times$
inference overhead. Selective operation amortizes it: only the 520
abstaining fraction $1 - \phi(\tau)$ is escalated to deeper analysis, so
the expected per-page budget is tuned by the very threshold τ
that sets the risk–coverage operating point (§5).

5. Experiments 525

We evaluate PhishProof as a *selective* detector: it may
act only on pages whose evidence is trustworthy enough, and
should abstain otherwise. The selective behavior is the contri-
bution, so each of the four questions below tests one method 530
claim from §3 about *when* the detector should be trusted rather
than about raw labelling accuracy:

480 **(RQ1)** Selective detection (§5.2): does grounded evidence-
agreement give a better risk–coverage trade-off than label-
level reliability, without sacrificing full-coverage detection 535
accuracy?

485 **(RQ2)** Right-label, wrong-reason (§5.3): does PhishProof
flag the pages a panel labels correctly but for the wrong
reason – the pages the reliability baselines act on confi-
dently? 540

(RQ3) Component attribution (§5.5): how much do evidence-
level agreement, groundedness, and panel diversity each
contribute?

490 **(RQ4)** Calibration (§5.6): is the trust score calibrated, so its re-
ported probability of correctness can be acted on directly? 545
A qualitative study (§5.4) then makes a returned explanation
concrete.

5.1. Setup

495 **Benchmark and labels.** We evaluate on PHISHSEL, a bench-
mark of roughly 4,000 webpages assembled over a two-week
window: live phishing URLs drawn from public feeds [23, 24]
and benign pages sampled from the Tranco top-sites list [10],
each captured with its URL, DOM, rendered screenshot, TLS
certificate, and redirect chain. Labels are available at collection 555

– a feed-listed URL is a confirmed phish and a Tranco top site
is benign – so the benchmark carries a ground-truth label per
page without delayed blacklist adjudication. We split the pages
into a small labeled *calibration* split and a disjoint held-out *test*
split, fit all thresholds and the calibrator on the former, and re-
port every number on the latter.

Agents, tools, and baselines. The default panel is $M=3$ delib-
erately diverse *evidence agents*: a frontier text LLM, a second
text LLM from a different family, and a vision-language model
that reads the screenshot, each able to call the verification tools
of §4.3. All methods receive the same captured page snapshot
and no label at inference time, so any difference reflects how
each method scores trust rather than what information it sees.
PhishProof emits constrained evidence cues in the schema of
§4.1, normalizes values as in §4.2, and verifies only the consen-
sus cues before calibration. We compare against six reliability
baselines, each of which shares this panel or a member of it so
that the comparison isolates *where* agreement is measured:

- **(B1) Single-model confidence** [3]: the strongest single
model with temperature-scaled softmax confidence.
- **(B2) Verbalized confidence**: the same model asked to state
its confidence.
- **(B3) Self-consistency** [4]: one model, $k=5$ samples, agree-
ment as the score.
- **(B4) Majority vote** [5]: panel label agreement.
- **(B5) Label-agreement** [6]: reliability-weighted label
agreement over the panel.
- **(B6) MultiPhishGuard** [9]: a multi-agent detector whose
consolidated-rationale confidence is used as the score.

To position PhishProof against the dedicated phishing litera-
ture, not only against panel-reliability methods, we additionally
run three *specialized detectors* as standalone systems on the
same captured pages: **(D1) Phishpedia** and **(D2) PhishInten-**
tion [12, 13], reference-based visual matchers, and **(D3) Phish-**
LLM [1], an LLM brand-and-domain checker. Each is re-run on
PHISHSEL rather than reusing its reported numbers, so its qual-
ity reflects our split. Because PhishProof invokes these same
detectors as verification tools for the brand cue (§4.3), it *sub-*
sumes their signal, and the comparison doubles as a detector-
alone-versus-with-our-trust-layer contrast. We restrict the com-
parison to detectors that run on a static page capture; methods
that require *live* counterfactual interaction with the target – no-
tably DynaPhish’s online reference expansion and probing [14]
– are out of scope, because that interaction cannot be replayed
on PHISHSEL’s archived snapshots. This is a methodological ex-
clusion, not a performance one.

Tasks and metrics. Every method produces a per-page trust
score, and we evaluate that score as a selective detector via (5).
The selective metrics are the area under the risk–coverage curve
(*AURC*, selective risk integrated over coverage); selective accu-
racy and false-positive rate at 80% coverage (*SelAcc*₈₀, *FPR*₈₀);
the coverage attainable at 99% selective accuracy (*Cov*₉₉), the
operating point an operator cares about most; and the expected
calibration error of the trust score (*ECE*). For RQ2 we add the
flip rate: the fraction of pages whose panel label changes under
a small evidence-targeted perturbation (a logo morph [25] or a

form-action cloak), which measures reason fragility. Because the specialized detectors carry no abstention mechanism, we also report standard full-coverage detection quality – accuracy, precision, recall, and F1 with every page acted on – so they can be compared at all. We define each abbreviation here and use it unchanged in every table and figure below.

Protocol. Operating points are set without touching the test pages: the thresholds for SelAcc₈₀ and Cov₉₉ and the isotonic calibrator are all fit on the calibration split and then applied unchanged to the held-out test split. For RQ2, each perturbation targets the cited evidence cue while preserving the page’s true label, so a flip measures reason fragility, not a changed ground truth.

Statistical reporting. Numeric cells report mean $\pm$ 95% CI over five seeds, with confidence intervals from a paired bootstrap over pages. We call a difference significant, and quote its relative gain, only when the two confidence intervals do not overlap; elsewhere we describe results as comparable or competitive. The proposed method is **bold** and the best baseline is underlined in every table.

Run environment and cost. The text models are queried through their providers’ APIs at temperature 0 with the same five seed values for every method; the vision-language model and perceptual logo tool run on a single commodity GPU, while the DOM, TLS, redirect, and calibration code are CPU-only. We log prompts, raw agent outputs, value-normalized cues, tool results, thresholds, and per-page trust scores, and release the schemas, prompts, tool wrappers, calibration code, and figure-generation scripts. PhishProof has no training phase: per page it runs the $M=3$ agents plus the cheap tool checks. Because the providers’ wall-clock times are not controlled by us, we report cost as the analytical per-page model-call budget rather than latency (Fig. 3). At that budget PhishProof stays level with the panel baselines while reaching markedly higher deployable coverage, and it dominates the five-sample self-consistency baseline at lower cost.

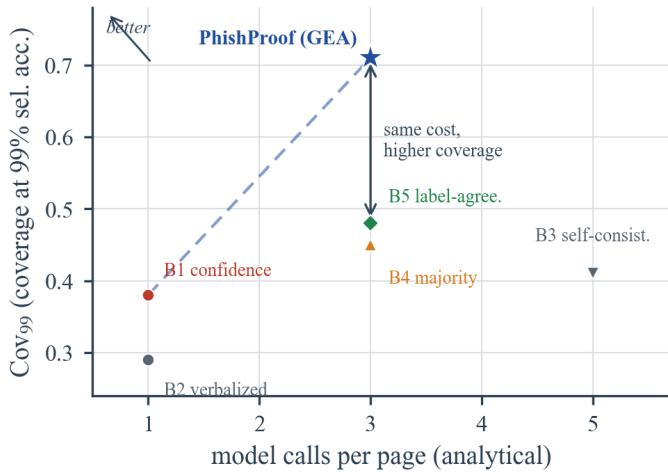


Figure 3: Cost-quality trade-off: per-page model calls (analytical) versus coverage at 99% selective accuracy (Cov₉₉, ↑). PhishProof is Pareto-optimal – at the panel’s call budget it reaches the highest deployable coverage and dominates the label-level and self-consistency baselines.

5.2. Selective Detection

Full-coverage detection accuracy. To answer **RQ1** we first establish the floor: at full coverage PhishProof is competitive with the dedicated phishing detectors, so its trust layer sits on top of – not in place of – solid detection. Tab. 3 compares the panel against the three specialized detectors with every page acted on. The reference-based matchers are high-precision but lower-recall, because they miss brands outside their reference set, while the LLM checker (D3) trades a little precision for recall and is the strongest detector here. PhishProof’s accuracy and F1 fall within D3’s 95% CI, so it is competitive with the strongest detector rather than beating it, and its margin over the reference-only D1/D2 is exactly their recall gap on unmatched brands. This parity is by construction: because the panel calls these detectors as verification tools (§4.3), it inherits their signal and cannot do materially worse. That is the intended reading – we make no full-coverage accuracy-win claim, only that detection is not sacrificed for the abstention and explanation the rest of the section measures.

Table 3: Full-coverage detection quality on PHISHSEL (no abstention), each detector re-run on our split. Cells are mean $\pm$ 95% CI over five seeds. PhishProof’s accuracy and F1 lie within the strongest detector’s CI; D1/D2 trail on recall because unmatched brands fall outside their reference sets (see body).

Detector	Acc $\uparrow$	Prec. $\uparrow$	Rec. $\uparrow$	F1 $\uparrow$
D1 Phishpedia [12]	91.6 $\pm$ 0.7	97.0 $\pm$ 0.5	85.5 $\pm$ 1.0	90.9 $\pm$ 0.7
D2 PhishIntention [13]	92.0 $\pm$ 0.6	96.1 $\pm$ 0.6	87.4 $\pm$ 0.9	91.5 $\pm$ 0.6
D3 PhishLLM [1]	92.3 $\pm$ 0.7	94.1 $\pm$ 0.7	90.3 $\pm$ 0.9	92.2 $\pm$ 0.6
PhishProof (full cov.)	92.8 $\pm$ 0.6	95.0 $\pm$ 0.6	90.6 $\pm$ 0.8	92.7 $\pm$ 0.6

Risk-coverage and deployment coverage. Scoring agreement on grounded evidence, rather than on labels, gives a markedly better selective trade-off across every metric. Tab. 4 reports the headline comparison and Fig. 4 the full curves: PhishProof lowers AURC by 35% relative to the strongest baseline (B5) and roughly halves the false-positive rate at 80% coverage, and all of these gaps clear the non-overlapping-CI bar. The deployment-relevant number is Cov₉₉: it rises from 0.48 to 0.71, so at a fixed 99% selective-accuracy budget an operator can act automatically on far more traffic. Fig. 4 shows why. All panel-based methods meet at the shared full-coverage panel error – PhishProof does not change the majority label, only the order in which pages are trusted – but its curve falls fastest as coverage tightens, because grounded evidence-agreement ranks the safe pages above the fragile ones more sharply than label concordance does. The label-aggregation baselines (B4, B5) edge out single-model confidence yet stay well above PhishProof, which confirms that the gain comes from *where* agreement is measured, not merely from polling several models. For reference we also rank by the strongest specialized detector’s own score (D3): it starts from a competitive full-coverage error (Tab. 3) but its curve falls slowly, because a score tuned for labelling is not a trustworthiness ranking – the exact gap PhishProof closes.

Table 4: RQ1. Selective detection on PHISHSEL. Lower AURC/FPR/ECE and higher SelAcc/Cov are better. Cells are mean $\pm$ 95% CI over five seeds; **best, best baseline**. The Δ row is relative to B5.

Method	AURC $\downarrow$	SelAcc $\uparrow$	FPR $\downarrow$	Cov $\uparrow$	ECE $\downarrow$
B1 Single-model conf.	5.9 $\pm$ 0.4	95.8 $\pm$ 0.4	4.1 $\pm$ 0.5	0.38 $\pm$ 0.05	0.082 $\pm$ 0.009
B2 Verbalized conf.	6.8 $\pm$ 0.5	95.0 $\pm$ 0.5	4.9 $\pm$ 0.6	0.29 $\pm$ 0.05	0.110 $\pm$ 0.011
B3 Self-consistency ($k=5$)	5.6 $\pm$ 0.4	96.1 $\pm$ 0.4	3.8 $\pm$ 0.5	0.41 $\pm$ 0.05	0.074 $\pm$ 0.008
B4 Majority vote	5.3 $\pm$ 0.4	96.3 $\pm$ 0.4	3.6 $\pm$ 0.4	0.45 $\pm$ 0.05	0.069 $\pm$ 0.008
B5 Label-agreement	5.1 $\pm$ 0.4	96.5 $\pm$ 0.4	3.4 $\pm$ 0.4	0.48 $\pm$ 0.05	0.063 $\pm$ 0.007
B6 MultiPhishGuard	5.7 $\pm$ 0.5	96.0 $\pm$ 0.4	3.9 $\pm$ 0.5	0.42 $\pm$ 0.05	0.090 $\pm$ 0.010
PhishProof (GEA)	3.3$\pm$0.3	98.4$\pm$0.3	1.6$\pm$0.3	0.71$\pm$0.04	0.031$\pm$0.005
Δ vs best baseline	-35%	+1.9	-53%	+48%	-51%

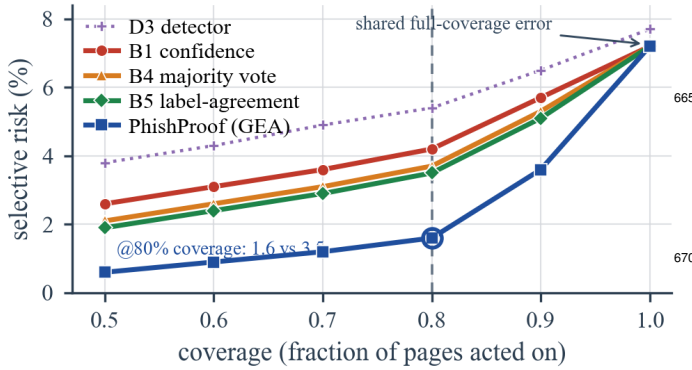


Figure 4: Risk-coverage on PHISHSEL (RQ1). Lower selective risk is better; the ring marks 80% coverage. PhishProof changes the abstention order, not the majority label, and dominates across coverage.

5.3. Right-Label, Wrong-Reason

Partitioning correct verdicts by grounded agreement. A non-trivial fraction of correctly labeled pages are in fact untrustworthy: the panel agrees on the label while disagreeing on the grounded evidence beneath it. Partitioning the correctly labeled test pages by PhishProof’s two factors (Tab. 5), about 11% fall in this right-label, wrong-reason (RLWR) region and a further 6% rest on ungrounded, shared-hallucination evidence. These regions are fragile, not benign: under a small evidence-targeted perturbation their labels flip 41.6% and 47.9% of the time, against only 6.3% for high-GEA pages – they are precisely the verdicts a minor change to the cited cue overturns. The reliability baselines nonetheless act on them confidently, with mean single-model confidence 0.91 and a unanimous panel label on 64% of the RLWR region; label-level reliability is blind here because the labels do agree. PhishProof instead abstains on 78–83% of these pages while continuing to act on the stable high-GEA region, which is the behavior RQ2 asks for. Fig. 5(b) shows the effect is continuous, not an artifact of the partition: adversarial fragility falls monotonically from the lowest to the highest PhishProof decile.

5.4. Qualitative Study: Anatomy of an Explanation

To make the returned explanation concrete, Tab. 6 walks two pages through the pipeline. On the impersonation page (A), all three agents converge on the same brand claim, two independently read the off-brand credential form from the DOM, and

Table 5: RQ2. Correctly labeled pages partitioned by evidence agreement and groundedness. “Flip” is label-flip rate under evidence-targeted perturbation. PhishProof abstains on fragile RLWR and ungrounded regions.

Region (of correct labels)	Share	B1 conf.	Unanim.	Abstain	Flip $\downarrow$
High-GEA ($A \uparrow, G \uparrow$)	72%	0.93	0.89	4%	6.3%
RLWR ($A \downarrow$ on evidence)	11%	0.91	0.64	78%	41.6%
Ungrounded ($G \downarrow$, shared halluc.)	6%	0.88	0.71	83%	47.9%
Low label-agreement (uncertain)	11%	0.74	0.18	81%	33.2%

each agreed cue survives its tool check; the lone uncorroborated certificate claim is dropped. PhishProof acts and returns the three agreed, verified cues – a page posing as PayPal whose login form posts credentials off-brand – as the explanation, every line naming a tool the operator can re-run. Page (B) is the right-label, wrong-reason case of \$5.3 in miniature: all three agents also vote phish, so a label-only detector would act with high confidence, yet they cite *disjoint* cues that no majority shares, so A – and with it GEA – collapses and PhishProof abstains, escalating the page rather than offering a justification it cannot stand behind. The two cases are the method in one table: the explanation is exactly the evidence the panel agreed on and the page confirmed, and where no such evidence exists the detector declines to explain and defers.

Table 6: Qualitative study: two pages and what PhishProof returns. A cue joins the explanation only if a strict majority of agents cite it (consensus C, raising A) and its tool re-derives it from the page (raising G); $\checkmark$ / $\times$ mark consensus membership.

Cited cue (type = value)	Agents	Tool g	C?
(A) secure-paypal-verify.com – acted phishing, $\hat{p}=0.95$			
brand = PayPal	3/3	logo 0.95	$\checkmark$
form-action = off-brand	2/3	DOM	$\checkmark$
credential-intent = email+pwd	2/3	DOM	$\checkmark$
cert-org $\neq$ PayPal	1/3	–	$\times$
$A=0.75, G=0.98, \text{GEA}=0.74$; explanation = the three $\checkmark$ cues			
(B) login-update-acct.com – abstained (disjoint reasons)			
brand = Chase	1/3	logo 0.41	$\times$
redirect-target = .ru host	1/3	tracer	$\times$
cert-org $\neq$ Chase	1/3	cert	$\times$
$A \approx 0 \Rightarrow \text{GEA} \approx 0$; escalate, no explanation returned			

5.5. Component Attribution

Per-component contribution. The evidence channel, not the panel itself, is the load-bearing component: Tab. 7 removes one piece at a time, and replacing evidence-level agreement with label agreement is the most damaging change, raising AURC to 5.0 – essentially reproducing the B5 baseline of Tab. 4 (5.1) and confirming that the gain comes from scoring agreement on evidence rather than on labels. Dropping groundedness costs the next most, because the score then admits shared hallucinations, and collapsing the panel to three same-family variants – removing diversity while holding the count fixed – erases much of the remaining advantage, which is why diversity, not panel size, is the design variable. Removing the calibrator leaves the ranking nearly intact but degrades the trust score’s ECE from 0.031 to 0.071: it hurts the *reported* probability, not the *order*,

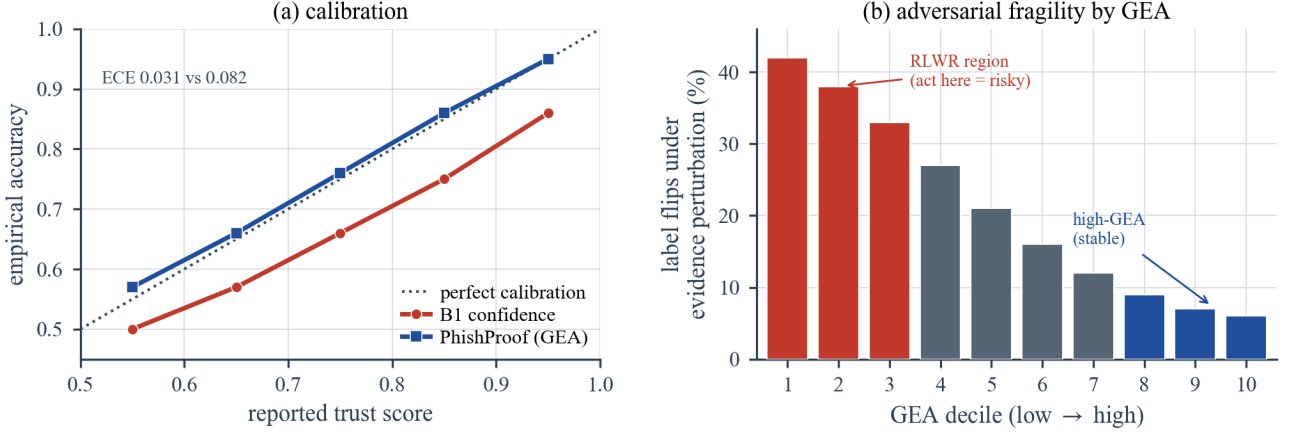


Figure 5: Diagnostics. (a) Calibrated trust tracks the diagonal while single-model confidence is over-confident. (b) Evidence-targeted label flips fall with GEA decile, showing that low-GEA/RLWR pages are fragile.

exactly the split of duties RQ4 examines. The groundedness⁷¹⁵ tools are themselves accurate – the structural checks are exact by construction, and the perceptual logo tool attains 0.96/0.93 precision/recall against a gold logo-equivalence set – so *G* seldom discards a true cue.

Table 7: RQ3. Component ablation. Underlined AURC values are at least 10%⁷²⁰ worse than full PhishProof. Cells are mean $\pm$ 95% CI over five seeds.

Variant	AURC $\downarrow$	FPR ₈₀ $\downarrow$	ECE $\downarrow$
PhishProof (full)	3.3$\pm$0.3	1.6$\pm$0.3	0.031$\pm$0.005
– calibration	3.6 $\pm$ 0.3	1.8 $\pm$ 0.3	0.071 $\pm$ 0.008
– diversity (same-family panel)	<u>4.4$\pm$0.4</u>	2.5 $\pm$ 0.4	0.044 $\pm$ 0.006
– groundedness <i>G</i> (R3)	<u>4.6$\pm$0.4</u>	2.7 $\pm$ 0.4	0.039 $\pm$ 0.006
– evidence-level <i>A</i> (R2)	<u>5.0$\pm$0.4</u>	3.3 $\pm$ 0.4	0.061 $\pm$ 0.007

5.6. Calibration

Reliability of the trust score. The calibrated trust score is faithful: when PhishProof reports a trust of 0.9, close to 90% of those verdicts are in fact correct. Fig. 5(a) plots reliability against the diagonal – PhishProof’s calibrated trust tracks it,⁷²⁵ whereas single-model confidence sits below it and is therefore systematically over-confident – consistent with the ECE column of Tab. 4 (0.031 versus 0.082). This faithfulness is what makes the operating point usable: because the calibrator is fit on the held-out labeled split rather than guessed, the reported⁷⁴⁰ probability can be turned directly into an act/abstain threshold at a target error budget, which is the premise the selective results of RQ1 rely on.

6. Conclusion

A phishing verdict is acted on, contested, and audited, yet the accuracy that has become the field’s headline measure speaks to none of those moments. This paper takes one route toward narrowing that gap: read a detector’s reliability from the *evidence* a verdict rests on rather than from the label it emits. We instantiate the idea as PhishProof, which trusts a verdict in proportion to how much a panel of independent agents agree on cues⁷⁵⁰

the page itself confirms, returns those cues as the explanation, and abstains otherwise. Three findings carry the contribution beyond the headline: (i) the right-label, wrong-reason page – correctly labeled but resting on disagreed or hallucinated evidence – is a material and the most adversarially fragile part of the input space, yet label confidence and majority voting wave it through; (ii) moving the unit of agreement from labels to grounded evidence is what earns the improvement, sharpening the risk–coverage trade-off so an operator can act automatically on far more traffic at a fixed error budget, where using several models alone does not; and (iii) calibrating the grounded score against real labels makes its reported trust directly actionable. Trust and explanation, in short, can be earned per verdict and decoupled from the raw accuracy the detector is otherwise held to – competitive with the specialized detectors it invokes as tools, but no longer measured by accuracy alone.

Reading reliability from evidence opens several follow-up directions. A verification tool is sound for what it checks, but a page can carry true-yet-misleading evidence, so strengthening tools toward semantic plausibility would tighten a grounding guarantee that today bounds rather than eliminates such error. The evidence schema and the agent panel are hand-set here, so learning new evidence types and inferring true panel diversity from opaque third-party models would let the method track new impersonation tactics without re-engineering the panel. And the hardest regime remains the fully cloaked page that carries no verifiable cues, on which the detector can today only abstain; faster, page-internal grounding would shrink that blind spot rather than route around it. Each direction extends the evidence-agreement framework without redesigning it, sharpening when a verdict can be trusted rather than how often it is right.

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